SOLUTION

SECTION–I

1. Ans. (C)

$$\int_{-T/2}^{3T/2} f(x)dx = 18 \Rightarrow \int_{0}^{2T} f(x)dx = 18 \Rightarrow \int_{0}^{T} f(x)dx = 9$$

$$\int_{-a}^{a+5T} f(x)dx = \int_{-a}^{a} f(x)dx + \int_{a}^{a+5T} f(x)dx$$

$$= 2\int_{0}^{a} f(x)dx + 5\int_{0}^{T} f(x)dx = 2 \times 3 + 5 \times 9 = 51$$

2. Ans. (C)

$$\int_{3/2}^{15} [\ell n[x]]dx = \int_{3/2}^{3} 0.dx + \int_{3}^{8} 1.dx + \int_{8}^{15} 2.dx = 0 + (8 - 3) + 2(15 - 8) = 19$$

3. Ans. (D)

$$\int_{-2}^{4} f(x)dx = \int_{-2}^{-1} f(x)dx + \int_{-1}^{0} f(x)dx + \int_{0}^{1} f(x)dx + \int_{1}^{2} f(x)dx + \int_{2}^{3} f(x)dx + \int_{3}^{4} f(x)dx$$

$$= 2^2 + 1^2 + 0 + 1^2 + 2^2 + 3^2 = 19$$

4. Ans. (B)

$$I = \int_{-1}^{1} \frac{x \tan^{-1} x}{e^{\tan^{-1} x} + 1} dx \quad \text{.....(i)}$$

$$\text{Also} I = \int_{-1}^{1} \frac{x \tan^{-1} x}{1 + e^{-\tan^{-1} x}} dx \quad \text{.....(ii)}$$

add (i) & (ii)

$$2I = \int_{-1}^{1} x \tan^{-1} x \, dx$$

$$I = \int_{0}^{1} x \tan^{-1} x = \frac{x^2}{2} \tan^{-1} x \Big|_0^1 - \frac{1}{2}\int_0^1 \frac{x^2}{x^2+1} dx$$

$$I = \frac{\pi}{8} - \frac{1}{2}\int_0^1 \frac{x^2+1-1}{x^2+1} = \frac{\pi}{8} - \frac{1}{2} + \frac{1}{2}\left(\tan^{-1} x\right)_0^1$$

$$I = \frac{\pi}{4} - \frac{1}{2}$$

5. Ans. (D)

$x^2 - 4x + 6 = A(x^2 - 2x + 4) + B(2x - 2) + C$

$\Rightarrow A = 1, B = -1, C = 0$

$$\Rightarrow \int \frac{x^2 - 4x + 6}{x^2 - 2x + 4} dx = \int A dx + B\int \frac{dt}{t} + 0$$

where $t = x^2 - 2x + 4$

$= x - \ell n|x^2 - 2x + 4| + C$

6. Ans. (A)

$$I = \int \frac{\left(1 - \frac{\sin^2 x}{\cos^2 x}\right)}{\frac{1}{\cos^2 x}} \frac{dx}{\sin 2x}$$

$$= \int \cot 2x dx = \frac{\ell n|\sin 2x|}{2} + C$$

7. Ans.(D)

$$I = \int e^x \left(\ell nx + x\ell nx + 1\right)dx$$

$$= \int e^x \left(x\ell nx + \left(1 + \ell nx\right)\right)dx \Rightarrow e^x f(x) + c$$

$I = e^x.x \ \ell n\, x + C$

$\Rightarrow \quad f(x) = e^x.x \ln x + C$

$f(1) = 0 \Rightarrow \quad C = 0$

$\Rightarrow \quad f(e) = e^{e+1}$

8. Ans. (A,B,C,D)

Let $I = \int_0^{\pi} |\sin \lambda x| dx, \ \lambda \in N$

put $\lambda x = t \quad \Rightarrow \lambda dx = dt$

$$\Rightarrow \quad I = \frac{1}{\lambda}\int_0^{\lambda\pi} |\sin t| \, dt = \int_0^{\pi} |\sin t| \, dt = 2$$

Using this

$$I_{(2)} = \int_0^{\pi} (1 - |\sin x|)dx = \pi - 2$$

$$I_{(3)} = \int_0^{\pi} (1 - |\sin x| + |\sin 2x|)dx$$

$$= \int_0^{\pi} 1 dx - \int_0^{\pi} |\sin x| \, dx + \int_0^{\pi} |\sin 2x| \, dx = \pi - 2 + 2 = \pi$$

Similarly $I_{(2008)} = \pi - 2$ & $I_{(2009)} = \pi$

Rankers Academy JEE

For More Material Join: @JEEAdvanced_2024

9. Ans. (A,B,C)

$$I_n = \int_0^1 (1-x^2)^n\,dx$$

$$= (1-x^2)^n \cdot x\Big|_0^1 + 2n\int_0^1 (1-x^2)^{n-1}\,x^2\,dx$$

$$= 2n\int_0^1 (1-x^2)^{n-1}\left\{1-(1-x^2)\right\}dx$$

$$I_n = 2nI_{n-1} - 2nI_n \Rightarrow I_n = \frac{2n}{2n+1}I_{n-1}$$

$$I_1 = \int_0^1 (1-x^2)dx = \frac{2}{3}$$

$$I_2 = \frac{4}{5}\cdot\frac{2}{3},\quad I_3 = \frac{6.4.2}{7.5.3}$$

$$\vdots$$

$$I_n = \frac{2.4.6.....2n}{3.5.7.....(2n+1)} = \frac{2^n n!}{3.5.7.....(2n+1)}$$

10. Ans.(A,B,C)

$$\int \frac{3e^x+5e^{-x}}{4e^x-5e^{-x}}dx$$

$$(3e^x+5e^{-x}) = P(4e^x-5e^{-x}) + Q(4e^x+5e^{-x})$$

$$4P+4Q=3\ \&\ 5Q-5P=5 \Rightarrow P=-\frac{1}{8},\ Q=\frac{7}{8}$$

$$\int\left(-\frac{1}{8}+\left(\frac{7}{8}\right)\frac{4e^x+5e^{-x}}{4e^x-5e^{-x}}\right)dx$$

$$= -\frac{x}{8}+\frac{7}{8}\ell n\left|4e^x-5e^{-x}\right|+C$$

$$= -\frac{x}{8}+\frac{7}{8}\ell n\left|4e^{2x}-5\right|-\frac{7}{8}x+C$$

$$= -x+\frac{7}{8}\ell n\left|4e^{2x}-5\right|+C$$

$A=-1,\ B=\frac{7}{8}$, C is an arbitrary constant.

11 Ans. (A,C,D)

$$f^3(x) = \int_0^x t\,f^2(t)\,dt$$

Diffentiate both sides

$$3f^2(x)f'(x) = xf^2(x)$$

$$\Rightarrow\quad f'(x) = \frac{x}{3} \qquad \text{As } x \ge 0$$

$\Rightarrow$ $f(x)$ is increasing

$$\Rightarrow\quad \boxed{f(x) = \frac{x^2}{6}} \text{ as } f(0)=0$$

SECTION–III

1. Ans. 2

$$\frac{f'(x)}{f(x)} = \frac{f'(-x)}{f(-x)}$$

Integrate both the sides

$\ell n f(x) = -\ell n f(-x) + \ell nc$

$f(x)\,f(-x) = c$

Put $x = 0 \Rightarrow c = 1$

$$f(x)\,f(-x) = 1$$

$$I = \int_{-2}^{2} \frac{dx}{1+f(x)} \qquad ....(1)$$

$$I = \int_{-2}^{2} \frac{dx}{(1+f(-x)}$$

$$I = \int_{-2}^{2} \frac{dx}{\left(1+\frac{1}{f(x)}\right)} = \int_{-2}^{2} \frac{f(x)dx}{1+f(x)} \qquad ...(2)$$

$(1) + (2) \Rightarrow 2I = 4 \Rightarrow I = 2$

2. Ans. 4

$$I = \int_{1/2}^{2} e^{|x-1/x|}dx \qquad ...(i)$$

Let $x = \frac{1}{t} \Rightarrow dx = -\frac{dt}{t^2}$

$$I = -\int_{2}^{1/2} \frac{1}{t^2} e^{\left|\frac{1}{t}-t\right|}dt$$

$$I = \int_{1/2}^{2} \frac{1}{t^2} e^{\left|t-\frac{1}{t}\right|}dt \qquad ...(ii)$$

adding equation (i), (ii), we get

$$2I = \int_{1/2}^{2}\left(1+\frac{1}{t^2}\right)e^{\left|t-\frac{1}{t}\right|}dt = \int_{1/2}^{1}\left(1+\frac{1}{t^2}\right)e^{-t+\frac{1}{t}}dt + \int_{1}^{2}\left(1+\frac{1}{t^2}\right)e^{t-\frac{1}{t}}dt$$

Put $e^{-t+\frac{1}{t}} = k$ & $e^{t-\frac{1}{t}} = m$

$$\left(e^{-t+\frac{1}{t}}\right)\left(-1-\frac{1}{t^2}\right)dt = dk \ \&\ \left(1+\frac{1}{t^2}\right)e^{\left(t-\frac{1}{t}\right)}dt = dm$$

$$-\int_{e^{3/2}}^{1} dk + \int_{1}^{e^{3/2}} dm = 2I$$

$$2I = 2\int_{1}^{e^{3/2}} dm = 2\left[e^{3/2}-1\right]$$

$$I = \left(e^{3/2}-1\right) = \lambda_1\left(e^{\lambda_2/2}-1\right)$$

$\lambda_1 = 1$ and $\lambda_2 = 3$

$\therefore\ \lambda_1 + \lambda_2 = 4$

Rankers Academy JEE

For More Material Join: @JEEAdvanced_2024

3. Ans. 4

$f(x) = \sin(\cos x) + \cos(\sin x)$ is periodic with period 2π

$\Rightarrow$ From Jack property

$$\Rightarrow I_1 = 50\int_0^{2\pi}((\sin(\cos x)+\cos(\sin x))dx$$

$$\Rightarrow I_1 = 100\int_0^{\pi}(\sin(\cos x)+\cos(\sin x)dx \quad .....(i)$$

(From queen property $\because f(x) = f(2\pi - x)$)

$$\Rightarrow I_1 = 100\int_0^{\pi}(-\sin(\cos x)+\cos(\sin x)dx \quad .....(ii)$$

(From king property)

Add (i) & (ii)

$$\Rightarrow 2I_1 = 200\int_0^{\pi}\cos(\sin x)dx$$

$$\Rightarrow I_1 = 100\int_0^{\pi}\cos(\sin x)dx$$

$$I_1 = 200\int_0^{\pi/2}\cos(\sin x)dx$$

(Queen property)

Put sinx = t

cosxdx = dt

$$\Rightarrow dx = \frac{dt}{\sqrt{1-t^2}} \quad \Rightarrow \quad I_1 = 200\int_0^1\frac{\cos t\,dt}{\sqrt{1-t^2}}$$

$$\Rightarrow I_1 = 200I_2 \quad \Rightarrow \quad \frac{I_1}{50I_2} = 4$$

4. Ans. 7

Limit of a sum

$$\Rightarrow \frac{r}{n} \equiv x \quad \& \quad \frac{1}{n} \equiv dx$$

$$L = \lim_{n\to\infty}\sum_{r=n}^{2n}\left(\frac{1}{\frac{r^3}{n^3}\sqrt{\frac{r^4}{n^4}+1}}\times\frac{1}{n}\right)$$

$$= \int_1^2\frac{dx}{x^3\sqrt{x^4+1}} = \int_1^2\frac{dx}{x^5\sqrt{1+\frac{1}{x^4}}}$$

Let $1+\frac{1}{x^4} = t^2$

$$\Rightarrow \frac{dx}{x^5} = -\frac{tdt}{2}$$

$$\Rightarrow L = -\frac{1}{2}\int_{\sqrt{2}}^{\sqrt{17}/4}\frac{tdt}{t} = -\frac{1}{2}\left(\frac{\sqrt{17}}{4}-\sqrt{2}\right)$$

$$= \frac{1}{\sqrt{2}}-\frac{\sqrt{17}}{8} = \sqrt{\frac{1}{2}}-\sqrt{\frac{17}{64}} \Rightarrow \frac{32b}{a}-10 = 7$$

5 Ans. 3

$$I = \int\left(x^{12}+x^8+x^4\right)\left(2x^8+3x^4+6\right)^{1/4}dx$$

$$= \int\left(x^{11}+x^7+x^3\right)\left(2x^{12}+3x^8+6x^4\right)^{1/4}dx$$

Put $2x^{12} + 3x^8 + 6x^4 = t$

$\Rightarrow 24(x^{11} + x^7 + x^3)dx = dt$

$$\Rightarrow I = \int\frac{t^{1/4}}{24}dt = \frac{t^{5/4}}{24.\frac{5}{4}}+C$$

$$= \frac{1}{30}(2x^{12}+3x^8+6x^4)^{5/4}+C$$

$\Rightarrow (60a + 1) = (2+1) = 3$

SECTION–IV

1. Ans. (A)$\rightarrow$(q), (B)$\rightarrow$(s), (C)$\rightarrow$(p), (D)$\rightarrow$(p)

(A) $$I = \int_4^{10}\frac{[x^2]dx}{[(14-x)^2]+[x^2]} \quad \text{--------- (i)}$$

$$I = \int_4^{10}\frac{[(14-x)^2]}{[x^2]+[(14-x)^2]}dx \quad \text{--------- (ii)}$$

add (i) & (ii)

$$2I = \int_4^{10}dx$$

$\Rightarrow 2I = 6 \Rightarrow I = 3$

(B) $$\int_{-1}^{2}\frac{|x|}{x}dx = \int_{-1}^{0}(-1)dx+\int_0^2(1)dx = 1$$

(C) $$\lim_{n\to\infty}\sum_{r=1}^{n}\frac{r^{99}}{n^{100}} = \int_0^1x^{99}dx = \left[\frac{x^{100}}{100}\right]_0^1 = \frac{1}{100}$$

(D) $$5050\int_{-1}^{1}\sqrt{x^{200}}dx = 5050\times2\int_0^1|x^{100}|dx$$

$$= 5050\times2\int_0^1x^{100}dx$$

$$= 10100\times\left[\frac{x^{101}}{101}\right]_0^1 = 100 = \frac{1}{\alpha}$$

$$\Rightarrow \alpha = \frac{1}{100}$$

Rankers Academy JEE

For More Material Join: @JEEAdvanced_2024

2. Ans. (A)→(P,T); (B)→(T); (C)→(P,T); (D)→(R)

(A) $\int \cos^5 x dx = \int (1-\sin^2 x)^2 \cos x dx$

Let $\sin x = t$

$\Rightarrow \cos x\, dx$

$= dt = \int (1-t^2)^2 dt = \int (1+t^4-2t^2)dt$

$= t - \frac{2t^3}{3} + \frac{t^5}{5} + c$

$= \sin x - \frac{2}{3}\sin^3 x + \frac{\sin^5 x}{5} + c$

$\Rightarrow n + 2m = 5 - \frac{3}{2} \times 2 = 2$

(B) $I = \int \frac{x dx}{(x-3)(x-5)}$

$\Rightarrow 2I = \int \frac{(5(x-3)-3(x-5))dx}{(x-3)(x-5)}$

$= \ln|x-5|^5 - \ln|x-3|^3$

$I = \ell n \left|\frac{(x-5)^{5/2}}{(x-3)^{3/2}}\right| + c$

$\Rightarrow a + b = 4$

(C) $\frac{dy}{dx} = \frac{f'(t)}{g'(t)}$

$\frac{d^2y}{dx^2} = \frac{(g'(t).f''(t) - f'(t).g''(t))}{(g'(t))^2} \cdot \frac{1}{g'(t)}$

Put $t = 1$

$\frac{d^2y}{dx^2} = \frac{2}{1} = 2$

(D) $L = \lim_{x \to 0^+} (-\ell nx)^{\frac{1}{\ell nx}}$

$e^{\lim_{x\to 0^+} \frac{\ell n(-\ell nx)}{\ell nx}}$

Applying L'Hospital rule

$L = e^{\lim_{x\to 0^+} \frac{1}{-\ell nx} \cdot \frac{1}{x} \cdot x} = e^0 = 1$

$\Rightarrow \ell nL = 0$

Rankers Academy JEE

For More Material Join: @JEEAdvanced_2024